

Multiple Response Analysis

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Multiple Response Frequencies

Load Libraries

```
library(knitr)
library(rmarkdown)
library(htmlTable)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(readxl)
```

Load Data

```
setwd("~/Desktop/projects/r-starter")
# Replace 'your_dataset.csv' with the actual dataset file name
df <- read_xlsx("./MR_Drugs.xlsx") # Ensure the dataset is in the working directory
df
```

```
## # A tibble: 972 x 27
##   id    sex city inco1 inco2 inco3 inco4 inco5 inco6 inco7 pinco1 pinco2
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1001     2     1     0     0     0     0     0     1     0     6     -1
## 2 1002     2     1     0     1     0     0     0     0     0     2     -1
## 3 1003     2     1     0     0     0     0     0     1     0     6     -1
## 4 1004     2     1     0     1     0     0     0     0     0     2     -1
## 5 1005     2     1     0     0     0     0     0     0     1     7     -1
## 6 1006     2     1     1     1     0     0     0     0     0     2     1
## 7 1007     2     1     0     1     0     0     0     0     0     2     -1
## 8 1008     2     1     0     1     0     0     0     0     0     2     -1
## 9 1009     2     1     1     1     0     0     0     0     0     2     1
## 10 1010     2     1     0     1     0     0     0     0     0     2     -1
## # i 962 more rows
```

```
## # i 15 more variables: pinco3 <dbl>, pinco4 <dbl>, pinco5 <dbl>, pinco6 <dbl>,
## #   sinco1 <chr>, sinco2 <chr>, sinco3 <chr>, sinco4 <chr>, sinco5 <chr>,
## #   sinco6 <chr>, crime1 <dbl>, crime2 <dbl>, crime3 <dbl>, crime4 <dbl>,
## #   crime5 <dbl>
```

Process Multiple Response Data

```
# Define multiple response columns
income_mr_cols <- c("inco1", "inco2", "inco3", "inco4", "inco5", "inco6", "inco7")

# Convert wide format to long format
income_long <- df %>%
  select(all_of(income_mr_cols)) %>%
  pivot_longer(cols = everything(), names_to = "Income_MR", values_to = "Response") %>%
  filter(Response == 1) # Keep only selected responses

# Compute frequencies
income_freq <- income_long %>%
  group_by(Income_MR) %>%
  summarise(N = n()) %>%
  mutate(Percent = round((N / sum(N)) * 100, 1))

# Count unique cases
total_cases <- n_distinct(df$id) # Ensure there is a unique respondent ID column

# Compute percent of cases
income_freq <- income_freq %>%
  mutate(Percent_Cases = round((N / total_cases) * 100, 1))

# Add Total row
income_freq <- income_freq %>%
  bind_rows(tibble(
    Income_MR = "Total",
    N = sum(income_freq$N),
    Percent = 100.0,
    Percent_Cases = round(sum(income_freq$Percent_Cases), 1)
  ))
```

Display Frequency Table with htmlTable

```
htmlTable(
  income_freq,
  rnames = FALSE, # Hide row names
  header = c("", "N", "Percent", "Percent_Cases"), # Correct column names
  cgroup = c("", "Responses", ""), # Group headers
  n.cgroup = c(1, 2, 1), # Specify how many columns each header spans
  align = "lccc", # Align text in columns
  css.table = "margin: 20px; padding: 10px; color: black; border: 1px solid black; width: 50%;", # Add
  css.cell = "padding: 15px;" # Add padding to each cell
)
```

Responses

N

Percent

Percent_Cases

inco1

226

12.8

23.3

inco2

607

34.5

62.4

inco3

293

16.6

30.1

inco4

50

2.8

5.1

inco5

82

4.7

8.4

inco6

151

8.6

15.5

inco7

352

20

36.2

Total

1761

100

181

Multiple Response Analysis

```
library(foreign)
setwd("/Users/admin/Desktop/projects/r-starter")
# Create the data for the second table: "My friends will think I'm stupid for not being able to cope wi
friends_spss <- read.spss("./Mandip\ -\ SAQ8.sav", to.data.frame = TRUE)

# Combine all responses into a single vector
all_responses <- unlist(friends_spss)

# Get unique values and their frequencies
unique_values <- unique(all_responses)
frequency <- table(all_responses)

# Calculate total number of responses
total_responses <- length(all_responses)

# Calculate percentage
percent <- round((frequency / total_responses) * 100, 1)

# Valid percent is the same as percent in this case (no missing values assumed)
valid_percent <- percent

# Calculate cumulative percent
cumulative_percent <- round(cumsum(percent), 1)
```

```

# Create the result data frame
result_table <- data.frame(
  Valid = names(frequency),
  Frequency = as.numeric(frequency),
  Percent = as.numeric(percent),
  "Valid Percent" = as.numeric(valid_percent),
  "Cumulative Percent" = as.numeric(cumulative_percent)
)

# Add a "Total" row
total_row <- data.frame(
  Valid = "Total",
  Frequency = total_responses,
  Percent = 100.0,
  "Valid Percent" = 100.0,
  "Cumulative Percent" = 100.0
)

# Combine the result table and the total row
final_result <- rbind(result_table, total_row)

# Generate HTML for the second table
htmlTable(
  final_result,
  caption = "My friends will think I'm stupid for not being able to cope with SPSS",
  rnames = FALSE,
  align = "lccc", # Align text in columns
  css.table = "margin: 20px; padding: 10px; color: black; border: 1px solid black; width:100%;", # Add
  css.cell = "padding: 15px;border: 1px solid black;" , # Add padding to each cell align = "lccc", #
  cell.header = "padding: 15px;" # Add padding to each cell
)

```

My friends will think I'm stupid for not being able to cope with SPSS

Valid

Frequency

Percent

Valid.Percent

Cumulative.Percent

Strongly agree

3770

18.3

18.3

18.3

Agree

8351

40.6

40.6

58.9
Neither
4978
24.2
24.2
83.1
Disagree
2670
13
13
96.1
Strongly disagree
695
3.4
3.4
99.5
Strongly Disagree
104
0.5
0.5
100
Total
20568
100
100
100